A2A sums for $\pi\pi$ scattering

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1 Zeroth order diagrams

1.1 Connected pieces

1.1.1 Direct diagram

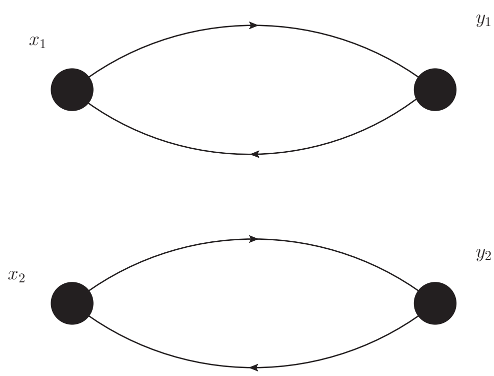

Figure 1: Direct contribution to I=2 scattering without EM corrections.

For the direct exchange diagram we have

$$C(x_1,x_2,y_1,y_2) = \mathrm{Tr}\left[\gamma_5 S_l(x_1,y_1)\gamma_5 S_l(y_1,x_1)\right] \cdot \mathrm{Tr}\left[\gamma_5 S_l(x_2,y_2)\gamma_5 S_l(y_2,x_2)\right]. \tag{1}$$

A2A factorization of the quark propagator

$$S(x,y) = \sum_{i=1}^{m} \left|v^i(x)\right\rangle \left\langle w^i(y)\right| \tag{2}$$

This becomes

$$\begin{aligned} C(x_1,x_2,y_1,y_2) = \mathrm{Tr}&\left[\gamma_5 \sum_{i_1=1}^{m} \left|v^{i_1}(x_1)\right\rangle \left\langle w^{i_1}(y_1)\right| \gamma_5 \sum_{i_2=1}^{m} \left|v^{i_2}(y_1)\right\rangle \left\langle w^{i_2}(x_1)\right|\right] \\ \cdot \mathrm{Tr}&\left[\gamma_5 \sum_{i_3=1}^{m} \left|v^{i_3}(x_2)\right\rangle \left\langle w^{i_3}(y_2)\right| \gamma_5 \sum_{i_4=1}^{m} \left|v^{i_4}(y_2)\right\rangle \left\langle w^{i_4}(x_2)\right|\right]. \end{aligned} \tag{3}$$

Take the summations out front and use the cyclic trace properties to get

$$C(x_1,x_2,y_1,y_2) = \sum_{i_1=1}^{m}\sum_{i_2=1}^{m} \mathrm{Tr}\left[\left\langle w^{i_2}(x_1)\right|\gamma_5\left|v^{i_1}(x_1)\right\rangle\left\langle w^{i_1}(y_1)\right|\gamma_5\left|v^{i_2}(y_1)\right\rangle\right] \cdot \sum_{i_3=1}^{m}\sum_{i_4=1}^{m} \mathrm{Tr}\left[\left\langle w^{i_4}(x_2)\right|\gamma_5\left|v^{i_3}(x_2)\right\rangle\left\langle w^{i_3}(y_2)\right|\gamma_5\left|v^{i_4}(y_2)\right\rangle\right]. \tag{4}$$

Introduce the pion meson fields

$$\Pi^{ij}(x) = \left\langle w^i(x)\right|\gamma^5\left|v^j(x)\right\rangle. \tag{5}$$

This yields

$$C(x_1,x_2,y_1,y_2) = \sum_{i_1=1}^{m}\sum_{i_2=1}^{m}\Pi^{i_2 i_1}(x_1)\Pi^{i_1 i_2}(y_1) \cdot \sum_{i_3=1}^{m}\sum_{i_4=1}^{m}\Pi^{i_4 i_3}(x_2)\Pi^{i_3 i_4}(y_2). \tag{6}$$

Where the traces have disappeared because the spinor indices have actually been contracted when we do $\langle w|\,\gamma_5\,|v\rangle$. We are typically interested in correlations which are functions of the time separation and moving pions. This can be accomplished by summing over the spatial coordinates $\vec{x}$ in $x = (\vec{x}, t_x)$ with the appropriate momentum phase

$$\Pi^{ij}(\vec{p},t) = V^{-1}\sum_{\vec{x}} e^{-i\vec{p}\cdot\vec{x}}\Pi^{ij}(x,t). \tag{7}$$

Doing this and plugging it in leaves us with

$$C(\vec{p}_{x1},\vec{p}_{x2},\vec{p}_{y1},\vec{p}_{y2};t_{x1},t_{x2},t_{y1},t_{y2}) = \sum_{i_1=1}^{m}\sum_{i_2=1}^{m}\Pi^{i_2 i_1}(\vec{p}_{x1},t_{x1})\Pi^{i_1 i_2}(\vec{p}_{y1},t_{y1}) \cdot \sum_{i_3=1}^{m}\sum_{i_4=1}^{m}\Pi^{i_4 i_3}(\vec{p}_{x2},t_{x2})\Pi^{i_3 i_4}(\vec{p}_{y2},t_{y2}). \tag{8}$$

Generically we are always going to be working in the center of mass frame, meaning we want $\vec{p}_{x2} = -\vec{p}_{x1}$ and $\vec{p}_{y2} = -\vec{p}_{y1}$. In the following we'll use $\vec{p}_x = \vec{p}_{x1} = -\vec{p}_{x2}$ and the same for y. Furthermore, we want to express the correlator in terms of the difference in times between the source pions $t_{x2} - t_{x1} = \Delta$, $t_{y2} - t_{y1} = \Delta$ and the overall times $t_{y1} - t_{x1} = \delta t$. Furthermore we call $t_x = t_{src}$. This yields

$$C(\vec{p}_x,\vec{p}_y,;t_{src},\Delta,\delta t) = \sum_{i_1=1}^{m}\sum_{i_2=1}^{m}\Pi^{i_2 i_1}(\vec{p}_x,t_{src})\Pi^{i_1 i_2}(\vec{p}_y,t_{src}+\delta t) \cdot \sum_{i_3=1}^{m}\sum_{i_4=1}^{m}\Pi^{i_4 i_3}(-\vec{p}_x,t_{src}+\Delta)\Pi^{i_3 i_4}(-\vec{p}_y,t_{src}+\delta t+\Delta). \tag{9}$$

Strip off the sums and make the matrix multiplications implicit:

$$C(\vec{p}_x,\vec{p}_y,;t_{src},\Delta,\delta t) = \mathrm{Tr}\Big[\Pi(\vec{p}_x,t_{src})\cdot\Pi(\vec{p}_y,t_{src}+\delta t)\Big]\,\mathrm{Tr}\Big[\cdot\,\Pi(-\vec{p}_x,t_{src}+\Delta)\cdot\Pi(-\vec{p}_y,t_{src}+\delta t+\Delta)\Big]. \tag{10}$$

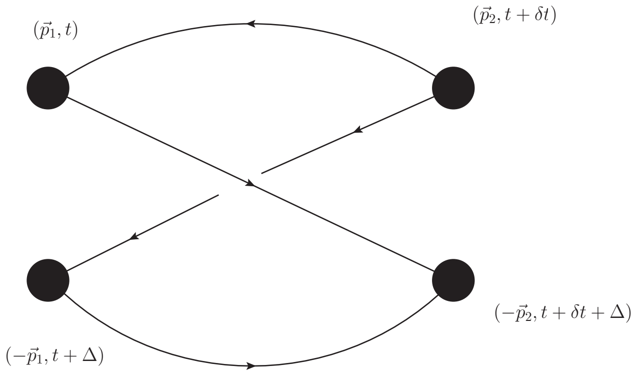

Figure 2: Cross contribution to I=2 scattering without EM corrections.

1.1.2 Cross diagram

$$C(\vec{p_1}, \vec{p_2}; t, \delta t) = \left\langle \mathrm{Tr}\left[\Pi(\vec{p_1}, t) \cdot \Pi(-\vec{p_2}, t+\delta t+\Delta)) \cdot \Pi(-\vec{p_1}, t+\Delta) \cdot \Pi(\vec{p_2}, t+\delta t)\right]\right\rangle \tag{11}$$

Where the meson field matrix is

$$\Pi(p,t) = \Pi_{ij}(p,t) \quad 0 \le i,j \le 2500; \tag{12}$$

1.2 Disconnected pieces

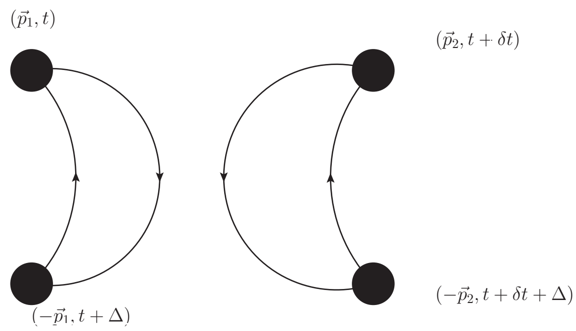

Figure 3: Disconnected contribution to I=0 scattering without EM corrections.

$$V(\vec{p_1}, \vec{p_2}; t, \delta t) = \left\langle \mathrm{Tr}\left[\Pi(\vec{p_1}, t) \cdot \Pi(-\vec{p}, t+\Delta)\right] \mathrm{Tr}\left[\Pi(\vec{p_2}, t+\delta t) \cdot \Pi(-\vec{p_2}, t+\delta t+\Delta)\right]\right\rangle \tag{13}$$

2 Direct exchange diagram

Consider Fig. 6. One propagator structure is

$$\mathrm{Tr}\{\gamma_5 S(x_1,y_1)\gamma_5 S(y_1,z_1)\gamma^\mu S(z_1,x_1)\} \cdot \mathrm{Tr}\{\gamma_5 S(x_2,y_2)\gamma_5 S(y_2,z_2)\gamma^\nu S(z_2,x_2)\} \tag{14}$$

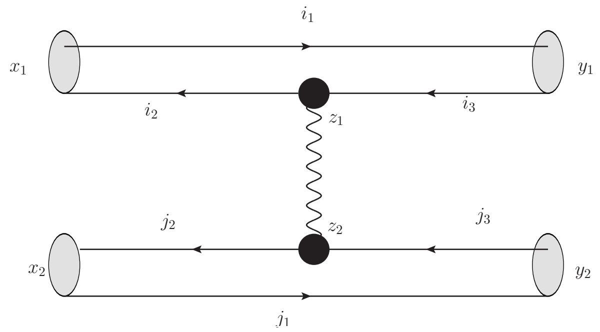

Figure 4: Radiative correction to direct exchange diagram. Labels $\{x, y, z\}$ are space-time indices and $\{i, j\}$ are A2A indices.

A2A factorization of the quark propagator

$$S(x, y) = \sum_{i=1}^{m} \left|v^i(x)\right\rangle \left\langle w^i(y)\right| \tag{15}$$

Pion meson field

$$\Pi^{ij}(x, t) = \left\langle w^i(x)\right| \gamma^5 \left|v^j(x)\right\rangle. \tag{16}$$

which can be projected to a non-zero momenta

$$\Pi^{ij}(\vec{p}, t) = V^{-1} \sum_{\vec{x}} e^{-i\vec{p}\cdot\vec{x}} \Pi^{ij}(x, t). \tag{17}$$

Or the obvious extension to a smeared source.

Simplifying Eq. (14) in terms of the pion meson fields, use momenta sources and sum over current insertion locations and we find

$$\begin{aligned} D(\vec{p}, \ldots) = \sum_{z_1, z_2} \Bigg[& \sum_{i_1, i_2, i_3} \Pi^{i_3 i_1}(\vec{p}, t_1) \Pi^{i_1 i_2}(\vec{p}, t_2) \left\langle w^{i_2}(z_1)\right| \gamma^\mu \left|v^{i_3}(z_1)\right\rangle \\ & \cdot \sum_{j_1, j_2, j_3} \Pi^{j_3 j_1}(\vec{p}, t_3) \Pi^{j_1 j_2}(\vec{p}, t_4) \left\langle w^{j_2}(z_2)\right| \gamma^\mu \left|v^{j_3}(z_2)\right\rangle \Bigg] \Delta_{\mu\nu}(z_1 - z_2). \end{aligned} \tag{18}$$

We have $m = 2700$ A2A vectors (2000 low modes and $\sim$700 high modes). We can take advantage of

$$y^{i_1}(z_1) = \sum_{i_2=1}^{m=2700} \Pi^{i_1 i_2}(\vec{p}, t) \left\langle w^{i_2}(z_2)\right|. \tag{19}$$

- Each vector operation is $O(m)$ which we for $\forall\ i_1, z_2$ implying that this evaluation is $O(m^2 V)$.
- Do this twice for each of the separate A2A index sums followed by a final $O(mV)$
- These two are only coupled by the final sum over current locations z_1 and z_2 which we can solve with FFT in $O(V \ln V)$

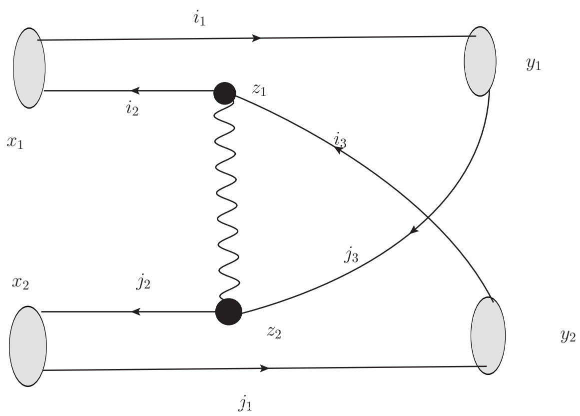

Figure 5: Radiative correction to cross exchange diagram. Labels $\{x, y, z\}$ are space-time indices and $\{i, j\}$ are A2A indices.

3 Cross exchange diagram

Consider one propagator structure

$$\mathrm{Tr}\{\gamma_5 S(x_1, y_1)\gamma_5 S(y_1, z_2)\gamma^\mu S(z_2, x_2)\gamma_5 S(x_2, y_2)\gamma_5 S(y_2, z_1)\gamma^\nu S(z_1, x_1)\} \tag{20}$$

Express in terms of the pion meson fields projected to certain momenta

$$\begin{aligned} \sum_{i_1,i_2,i_3} \sum_{j_1,j_2,j_3} \Pi^{i_1 i_2}(p_1)\Pi^{i_1 j_3}(p_1) \left\langle w^{j_3}(z_2)\right| \gamma^\mu \left|v^{j_2}(z_2)\right\rangle \\ \cdot \Pi^{j_2 j_1}(p_2)\Pi^{j_1 i_3}(p_2) \left\langle w^{i_3}(z_1)\right| \gamma^\nu \left|v^{i_2}(z_1)\right\rangle . \end{aligned} \tag{21}$$

Again we can use a series of vector-matrix operations of $O(m^2 V)$

$$\begin{aligned} \left\langle A^{i_2}(z_2)\right| &= \sum_{j_3 i_1} \Pi^{i_1 i_2}(p_1)\Pi^{i_1 j_3}(p_1) \left\langle w^{j_3}(z_2)\right| \\ \left|B^{i_3}(z_2)\right\rangle &= \sum_{j_2,j_1} \left|v^{j_2}\right\rangle \Pi^{j_2 j_1}(p_2)\Pi^{j_1 i_3}(p_2) \end{aligned} \tag{22}$$

Then we have

$$\sum_{z_1,z_2} \sum_{i_2,i_3} \left\langle A^{i_2}(z_2)\right| \gamma^\mu \left|B^{i_3}(z_2)\right\rangle \left\langle w^{i_3}(z_1)\right| \gamma^\nu \left|v^{i_2}(z_1)\right\rangle .\Delta_{\mu\nu}(z_1 - z_2) \tag{23}$$

It looks like the most straightforward way to do this might be to combine spatial indices with a FFT $\forall\ i_2 i_3$ implying $O(M^2 V \ln V)$. For these types of mixed-index contraction we use the FFT strategy 2, see Appendix C.

4 Self energy diagrams

Diagrams like these have propagator structures

$$\mathrm{Tr}\left[\gamma_\mu S_l(x1, x2)\gamma_\nu S_l(x2, snk1)\gamma_5 S_l(snk1, src1)\gamma_5 S_l(src1, x1)\right] \cdot \mathrm{Tr}\left[\gamma_5 S_l(src2, snk2)\gamma_5 S_l(snk2, src2)\right]. \tag{24}$$

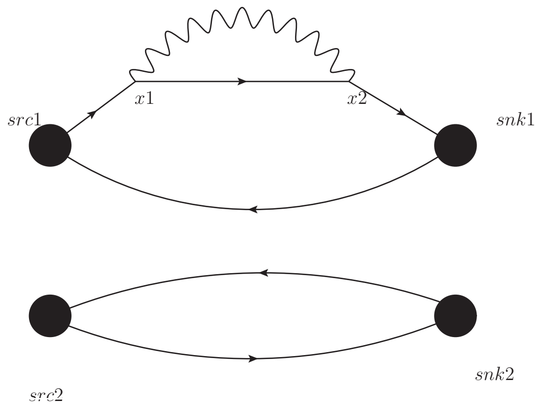

Figure 6: Self-energy type contribution to I=2 scattering.

If we simplify these into pion meson fields and A2A vectors we'll find something like (just consider the complicated piece)

$$\sum_{i_1,i_2,i_3.i_4} \left\langle w^{i_4}(x_1)\right| \gamma_\mu \left|v^{i_1}(x_1)\right\rangle \left\langle w^{i_1}(x_2)\right| \gamma_\mu \left|v^{i_2}(x_2)\right\rangle \Delta_{\mu\nu}(x_1 - x_2) \cdot \Pi^{i_2 i_3}(snk1)\Pi^{i_3 i_4}(src1). \tag{25}$$

Use the Matrix-vector operations

$$\left\langle z^{i_3}(x_1)\right| = \sum_{i_4} \Pi^{i_3 i_4}(src1) \left\langle w^{i_4}(x_1)\right|, \tag{26}$$

$$\left|y^{i_3}(x_2)\right\rangle = \sum_{i_2} \left|v^{i_2}(x_2)\right\rangle \Pi^{i_2 i_3}(snk1), \tag{27}$$

And this can be simplified to

$$\sum_{i_1,i_2}^{m=2700} \left\langle z^{i_3}(x_1)\right| \gamma_\mu \left|v^{i_1}(x_1)\right\rangle \left\langle w^{i_1}(x_2)\right| \gamma_\nu \left|y^{i_3}(x_2)\right\rangle \Delta_{\mu\nu}(x_1 - x_2) \tag{28}$$

Using the FFT we should be able to evaluate this with $O(m^2 V \ln V)$ operations, as with the cross exchange diagram.

5 Quark mass differences

These diagrams presumably don't need much fancy machinery and can be computed much in the same way as normal pion meson fields.

A FFT and convolution

Reduce sums that are naively $O(V^2)$

$$C = \sum_{x,y} \phi(x)\psi(y)\Delta(x-y), \tag{29}$$

To $O(V \ln V)$. Translational invariance allows us to diagonalize this inner product in momentum space

$$C = \sum_k \tilde{\phi}(-k)\tilde{\Delta}(k)\tilde{\psi}(k). \tag{30}$$

Workflow to actually accomplish this

1. $\tilde{\psi}(k) = FFT(\psi(x))$.
2. We know the explicit form of photon propagator $\Delta(k)$ in momentum space (Feynman or Coulomb gauge).
3. Site-wise product $\tilde{f}(k) = \tilde{\psi}(k) \cdot \Delta(k)$.
4. $f(x) = IFFT(\tilde{f}(k))$
5. $C = \sum_x \phi(x) \cdot f(x)$

NOTE: Grid/GPT FFT conventions include volume normalization on IFFT so I don't add them.

B FFT strategy type 1

These are the unmixed A2A index contractions such as

$$\sum_{z_1,z_2} \sum_{i,j} \left\langle A^i(z_2)\right| \gamma^\mu \left|B^i(z_2)\right\rangle \left\langle w^j(z_1)\right| \gamma^\nu \left|v^j(z_1)\right\rangle .\Delta_{\mu\nu}(z_1 - z_2) \tag{31}$$

The strategy for these types of diagrams is

```
for i:
    for mu:
        C_mu += GPU_local_inner_prod( <A_i (z1)| gamma_mu |B_j (z1)> )

for j:
    for nu:
        g_nu += GPU_local_inner_prod( <w_j (z2)| gamma_nu| v_j (z2)> )


for nu:
    # Kg = tilde(g)
    Kg_nu = FFT(g_nu)
    for mu:
        Kg_phtn_mu += Kg_nu * phtn_prop_mu_nu

for mu:
    g_phtn_mu = IFFT(Kg_phtn_mu)

    RESULT += GPU_site_sum( C_mu (z1) * g_phtn_mu (z1))
```

C FFT strategy type 2

These are the mixed A2A index contractions such as

$$\sum_{z_1,z_2}\sum_{i_2,i_3}\left\langle A^{i_2}(z_2)\right|\gamma^{\mu}\left|B^{i_3}(z_2)\right\rangle\left\langle w^{i_3}(z_1)\right|\gamma^{\nu}\left|v^{i_2}(z_1)\right\rangle.\Delta_{\mu\nu}(z_1-z_2) \tag{32}$$

We're saved from $O(m^3)$ by the trace on i_2. Workflow is as follows

```
for i_2:
    for i_3:
        for nu:
            g_i3i2_nu = GPU_local_inner_prod(<w_i3 (z2)| gamma_nu |v_i2 (z2)> )

            # Kg = tilde(g)
            Kg_i3i2_nu(k) = FFT(g_i2i3_nu)

            for mu:
                KG_i3i2_mu += GPU_local_inner_prod( Kg_i2i3_nu * phtn_prop_mu_nu )

        for mu:
            G_i3i2_mu = IFFT(KG_i2i3_mu)

            f_i2i3_mu = GPU_local_inner_prod( <A_i2 (z2)| gamma_mu |B_i3 (z2)> )

            RESULT += GPU_site_sum( G_i3i2_mu * f_i2i3_mu)
```